Title: Study of Digital to Analog Conversion using MATLAB

Objectives:

This experiment is designed to understand the use of MATLAB for solving communication engineering problems and to develop a basic understanding of digital-to-analog conversion techniques such as ASK, FSK, and PSK using MATLAB for simulation and analysis.

Theory:

A digital signal is a signal that represents data as a sequence of discrete values; at any given time, it can only take on one of a finite number of values. An analog signal is any continuous signal for which the time varying feature of the signal is a representation of some other time varying quantity. The following techniques can be used for Digital to Analog Conversion:

1. ASK (Amplitude Shift keying):

Amplitude Shift Keying is a technique in which carrier signal is analog and data to be modulated are digital. The amplitude of analog carrier signal is modified to reflect binary data. The binary signal when modulated gives a zero value when the binary data represents 0 while gives the carrier output when data is 1. The frequency and phase of the carrier signal remain constant.

2. FSK (Frequency Shift keying):

In this modulation the frequency of analog carrier signal is modified to reflect binary data. The output of a frequency shift keying modulated wave is high in frequency for a binary high input and is low in frequency for a binary low input. The amplitude and phase of the carrier signal remain constant.

3. PSK (Phase Shift keying):

In this modulation the phase of the analog carrier signal is modified to reflect binary data. The amplitude and frequency of the carrier signal remain constant.

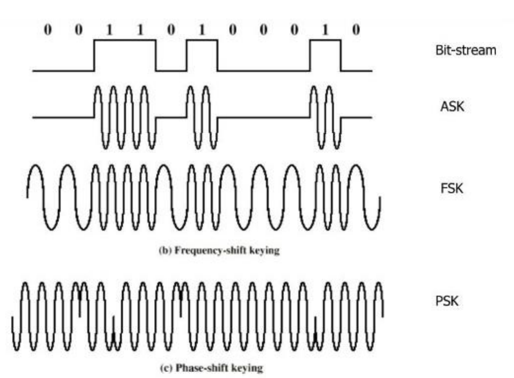


Fig: Implementation of Binary ASK, FSK and PSK

- **QPSK (Quadrature Phase Shift Keying):** Quadrature Phase Shift Keying is a digital modulation technique that improves upon Binary Phase Shift Keying by transmitting two bits per symbol instead of one. To achieve this, the input bitstream is divided into two parallel streams: one modulates the in-phase (I) carrier, while the other modulates the quadrature (Q) carrier, which is shifted by 90 degrees. These two carriers are then combined to produce a signal with four distinct phase states, typically at ±45° and ±135°. Each phase corresponds to a unique two-bit combination, allowing QPSK to double the data rate compared to BPSK without requiring additional bandwidth.

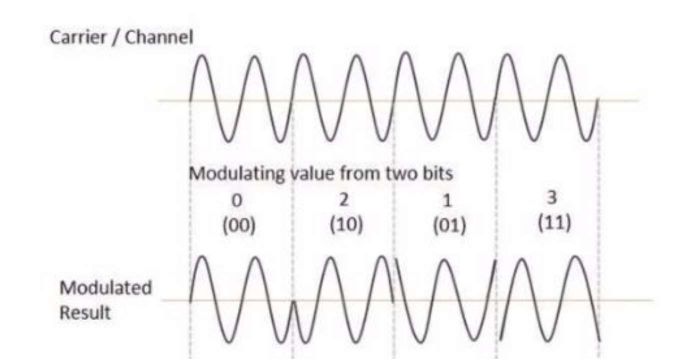


Fig: Implementation of Binary QPSK

Code:

8-ASK:

```
close all;
clc;
% Task: 8-ASK, 8-FSK, 8-PSK for E=9, F=7, G=2
% 24-bit stream from ASCII:
% '9' = 00111001
% '7' = 00110111
% '2' = 00110010
x = [0 0 1 1 1 0 0 1  0 0 1 1 0 1 1 1  0 0 1 1 0 0 1 0];
signal_count = 0;
nx=length(x);
f1=4;
i=1;

while i <= nx-2   % process 3 bits at a time

    t = (signal_count):0.001:(signal_count+1);
    % Take 3 bits
    bits = x(i:i+2);
    signal = bits(1)*4 + bits(2)*2 + bits(3); % Convert 3-bit binary to decimal (0-7)

    % 8-ASK
    % Amplitude: 0V to 7V
    ask = signal * sin(2*pi*f1*t);
```

```
subplot(3,1,1);
plot(t,ask);
hold on;
grid on;
axis([0 8 -8 8]);
title('8-ASK');
```

8-FSK:

```
% 8-FSK
% Frequency: 1 Hz to 8 Hz
f = signal + 1;
fsk = sin(2*pi*f*t);
```

```
subplot(3,1,2);
plot(t,fsk);
hold on;
grid on;
axis([0 8 -1 1]);
title('8-FSK');
```

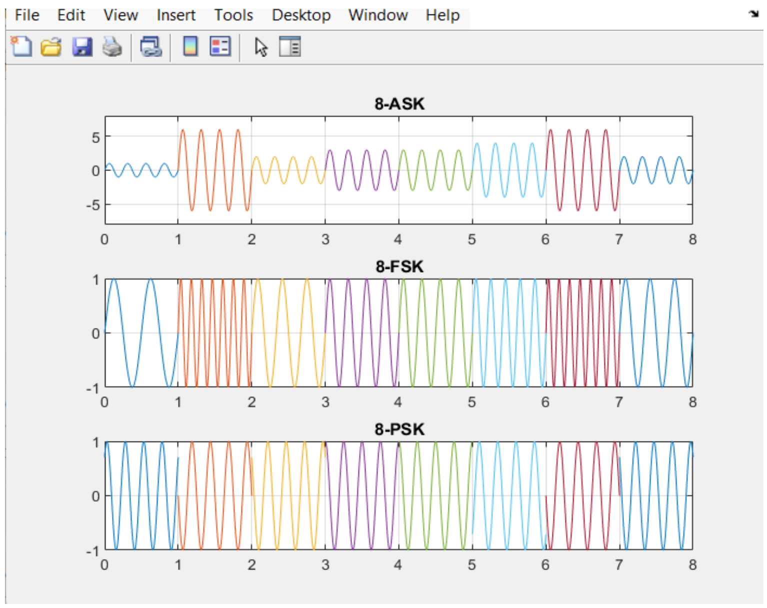
8-PSK:

```
% 8-PSK
phases = [0, pi/4, 3*pi/4, pi/2, -pi/4, -pi/2, pi, -3*pi/4];
psk = sin(2*pi*f1*t + phases(signal+1));
```

```
    subplot(3,1,3);
    plot(t,psk);
    hold on;
    grid on;
    axis([0 8 -1 1]);
    title('8-PSK');

    i = i + 3;  % move to next 3-bit symbol
   signal_count = signal_count + 1;
end
```

Simulation figure:



Result analysis/discussion:

The objective of this experiment was to study and implement different digital modulation techniques such as Amplitude Shift Keying (ASK). Frequency Shift Keying (FSK) and Phase Shift Keying (PSK) using. In the experiment, a binary sequence was used as input and three modulation schemes were applied. For (ASK), the carrier amplitude was present for bit '1' and absent for bit '0'. In FSK, the carrier frequency shifted between two values depending on the input bits. For PSK, the carrier phase was shifted by 180° for bit '0' and remained unchanged for bit '1'. The simulated results matched the theoretical expectations of digital modulating systems.

Conclusion:

The experiment successfully demonstrated ASK, FSK and PSK modulation using MATLAB simulation. It confirmed theoretical principles by accurately representing binary data through modulation. The work highlighted the importance of proper parameter selection, since poor choices reduce signal efficiency and clarity.

References:

[1] MathWorks, MATLAB Documentation, Natick, MA, USA: MathWorks. [Online]. Available: https://www.mathworks.com/help/

[2] Stormy Attaway, MATLAB: A Practical Introduction to Programming and Problem Solving, 6th ed. Boston, MA, USA: Butterworth-Heinemann, 2019